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XAI-DR-GAN: A topology-aware explainable GAN framework for interpretable diabetic retinopathy grading

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ABSTRACT

Diabetic Retinopathy (DR) is a progressive microvascular complication of diabetes characterized by biochemical and structural alterations in the retinal vasculature. Accurate automated analysis of these retinal changes is essential for early detection and grading of disease severity. Despite recent advances in deep learning, challenges such as data imbalance, imaging heterogeneity, and limited interpretability continue to hinder clinical integration. In this work, we propose XAI-DR-GAN, an explainable, topology-aware generative adversarial network designed for interpretable diabetic retinopathy grading and topology-preserving retinal augmentation. The framework integrates class-conditional generation, vascular topology preservation, and lesion-focused saliency alignment to synthesize anatomically plausible samples and enhance classification robustness. A Grad-CAM-based lesion alignment mechanism provides interpretable decision maps correlated with pathological biomarkers, while contrastive objectives facilitate domain adaptation across heterogeneous retinal datasets. Comprehensive experiments conducted on EyePACS, Messidor, APTOS, DDR, and DIARETDB1 datasets demonstrate that XAI-DR-GAN demonstrates improved and competitive performance compared to recent state-of-the-art methods in grading accuracy, interpretability alignment, and generative quality (FID, PSNR). These results demonstrate the effectiveness of topology-aware explainable generative learning for reliable, transparent, and data-efficient diabetic retinopathy screening across heterogeneous retinal imaging datasets

1. Introduction

Diabetic Retinopathy (DR) is a progressive microvascular complication of diabetes and remains one of the foremost causes of preventable blindness worldwide. The disease originates from a complex interplay of biochemical and cellular mechanisms, including pericyte apoptosis, endothelial cell damage, accumulation of advanced glycation end products (AGEs), oxidative stress, and chronic inflammation. These molecular and cellular alterations disrupt the integrity of retinal capillaries, leading to microaneurysms, vascular leakage, and ischemia-induced neovascularization. Such cascades ultimately result in irreversible damage to the retinal structure and visual function. Automated retinal analysis of these pathological changes has emerged as an essential approach for understanding disease dynamics, enabling predictive diagnosis, and supporting personalized treatment planning. With the increasing availability of low-cost retinal imaging devices—such as smartphone-based fundus cameras and portable optical coherence tomography (OCT) scanners—there is a growing need for intelligent, automated systems capable of mapping visible structural biomarkers to underlying biological processes. These computational frameworks must achieve not only high diagnostic performance but also biological interpretability, ensuring that model predictions align with known pathophysiological patterns through explainable AI (XAI) mechanisms [1].

Despite remarkable progress in deep learning-based DR analysis [2], two persistent challenges hinder clinical translation and integration into biomedical workflows. First, DR datasets exhibit significant class imbalance, where advanced yet biologically critical stages such as

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proliferative DR are underrepresented. This imbalance skews the optimization landscape, leading to biased learning of retinal patterns and poor generalization across disease stages. Second, conventional DR classification pipelines primarily emphasize image-level accuracy without incorporating biological interpretability. Establishing a computational link between imaging features and microvascular pathology is essential for deriving mechanistic insights and enhancing clinician trust. These limitations are further compounded in low-resource and decentralized environments, where imaging artifacts, variations in acquisition protocols, and biological heterogeneity introduce additional uncertainty in both data and model performance.

Existing literature has progressively advanced computational modeling of DR through multiple dimensions. Architecturally, convolutional neural network (CNN)–based systems such as DeepRetinaNet [3] and cascaded U-Net-ResNet hybrids [4] have enhanced the extraction of vessel morphology and lesion-specific patterns. More recent designs, such as C2x-FNet [5] and CFHAN [6], have achieved fine-grained DR grading through cross-feature fusion and hierarchical attention mechanisms that emphasize local–global contextual dependencies [7]. To address class imbalance, several studies have explored hybrid methods integrating evolutionary search techniques (e.g., Q-learning, particle swarm optimization, or genetic algorithms) with GAN-based augmentation strategies [8], enabling synthetic sample generation that complements scarce pathological classes. Similarly, hybrid CNN-Inception-ResNet architectures [9] have demonstrated strong robustness to variations in imaging modalities, while federated learning frameworks [10] have facilitated privacy-preserving collaboration across distributed medical centers. Furthermore, IoMT-based deployments [11] have contributed to near-real-time DR screening by integrating embedded AI inference modules for edge healthcare devices [12].

Explainability has become a cornerstone of computational biology and biomedical AI [13], bridging the gap between algorithmic decisions and biological reasoning. Attention-guided CNNs [14], image-quality-aware enhancement networks [15], and saliency-driven lesion localization modules have improved biological interpretability by linking network activation regions to pathological biomarkers such as microaneurysms, hemorrhages, and hard exudates. Recent research on robustness and generalization—such as exposure perturbation analysis [16] and domain adaptation studies [17,18]—has provided valuable insights into how models react to imaging noise and population diversity. Structural reasoning paradigms, including vascular-graph graph neural networks (GNNs) [19] and contrastive representation learning approaches like KMoCoL [20], have incorporated topological and semantic constraints to model retinal vasculature more faithfully. Additionally, advances in biomedical imaging, including vessel reconstruction using MaskVSC [21], elastography-based OCT imaging [22, 23], and multimodal registration frameworks [24], have expanded the diversity and richness of biological data available for computational analysis, facilitating integrative disease modeling [25].

Nevertheless, an end-to-end framework that jointly addresses biological imbalance, imaging variability, and intrinsic interpretability remains largely underdeveloped. Most existing GAN-based augmentation models assume high-quality clinical images and fail to preserve the fine vascular topology that encodes essential biological information. Furthermore, current XAI approaches are often applied post hoc and decoupled from model training [26], limiting their ability to influence the generation and decision-making processes. The absence of such unified modeling frameworks restricts both biological plausibility and clinical reliability, underscoring the need for integrative and explainable computational approaches.

Unlike prior GAN-based DR approaches such as DR-GAN [27] and EHO-Q-LGAN [8], which mainly emphasize adversarial retinal image generation or imbalance handling, the proposed XAI-DR-GAN focuses on a unified topology-aware and explainability-guided optimization strategy for DR grading. Specifically, the proposed framework jointly enforces vascular graph consistency through vessel-edge Wasserstein regularization, lesion-focused Grad-CAM [28] alignment, semantic contrastive consistency through the InfoNCE objective [29], and imbalance-aware classification within a single end-to-end training pipeline. Therefore, the contribution of this work lies in the coordinated integration of anatomical topology preservation [30], biologically consistent interpretability, and minority-grade augmentation for retinal disease grading, rather than in the isolated use of conditional GANs (cGANs) [31], focal loss [32], InfoNCE [29], or Grad-CAM [28] as independent components.

To address these limitations, we propose *XAI-DR-GAN*—an end-to-end, topology-aware, and explainable generative adversarial network designed for computational modeling and grading of DR progression. The proposed system integrates vascular topology constraints, semantic lesion priors, and explainability objectives within a unified optimization framework. By enforcing anatomical and functional consistency during training, the model produces biologically coherent synthetic images that aid both classifier balance and clinician interpretability. This work demonstrates the broader potential of explainable generative learning for enhancing retinal representation learning, biologically consistent interpretability, and reliable diabetic retinopathy grading within biomedical imaging applications.

The primary contributions of this work are summarized as follows:

1. We propose a topology-aware explainable GAN framework in which a class-conditional generator synthesizes anatomically coherent minority-class retinal images while preserving vascular continuity and lesion-aware structural consistency through a Wasserstein-based edge-set topology constraint for diabetic retinopathy augmentation and grading.
2. We introduce an interpretability-guided learning strategy that aligns Grad-CAM saliency responses with clinically relevant lesion regions, thereby improving biologically consistent decision-making and lesion-focused retinal analysis.
3. Extensive experiments across five benchmark retinal datasets demonstrate that the proposed framework improves balanced accuracy, QWK, interpretability alignment, robustness, and generative fidelity compared with recent state-of-the-art methods.

The remainder of this manuscript is organized as follows: Section 2 presents the proposed methodology. Section 3 details the datasets, evaluation metrics, implementation settings, and comparison with state-of-the-art methods. Section 4 reports ablation study results, and Section 5 concludes the study with potential future research directions. Additionally, Table 1 summarizes the abbreviations used throughout the manuscript for improved readability and clarity.

2. Proposed methodology

We propose **XAI-DR-GAN**, an explainable, topology-aware, and adaptive generative framework for diabetic retinopathy (DR) grading that unifies data augmentation, biological interpretability, and structural preservation within a single optimization paradigm. The framework is built upon a class-conditional generative adversarial network (cGAN) that models the conditional distribution of retinal fundus images given their corresponding disease grades. Unlike conventional augmentation techniques that rely solely on pixel-level realism, XAI-DR-GAN explicitly encodes vascular topology and lesion semantics to ensure that synthesized samples remain anatomically valid and biologically meaningful. The proposed design integrates a set of mathematically grounded regularization objectives, jointly optimized to satisfy the four key requirements for clinical-grade DR synthesis and classification: (i) *anatomical realism* to preserve retinal microvascular structure and lesion geometry, (ii) *class balance* to counteract data skew across disease stages, (iii) *semantic consistency* to maintain alignment between generated features and real pathological signatures, and (iv) *interpretability* to ensure transparent and biologically verifiable decision-making.

Table 1
List of abbreviations used in the manuscript.

Abbreviation	Expanded form
DR	Diabetic Retinopathy
XAI	Explainable Artificial Intelligence
GAN	Generative Adversarial Network
XAI-DR-GAN	Explainable Diabetic Retinopathy Generative Adversarial Network
CNN	Convolutional Neural Network
ViT	Vision Transformer
GNN	Graph Neural Network
IoMT	Internet of Medical Things
OCT	Optical Coherence Tomography
Grad-CAM	Gradient-weighted Class Activation Mapping
cGAN	Conditional Generative Adversarial Network
EMA	Exponential Moving Average
IoU	Intersection over Union
QWK	Quadratic Weighted Kappa
FID	Fréchet Inception Distance
PSNR	Peak Signal-to-Noise Ratio
IQA	Image Quality Assessment
APTOS	Asia Pacific Tele-Ophthalmology Society
DDR	Deep Diabetic Retinopathy Dataset
DIARETDB1	Diabetic Retinopathy Database 1
SOTA	State-of-the-Art
InfoNCE	Information Noise Contrastive Estimation
AGEs	Advanced Glycation End Products
RTX	Ray Tracing Texel eXtreme (NVIDIA GPU Series)
CUDA	Compute Unified Device Architecture
CNN–ViT	Convolutional Neural Network – Vision Transformer
Bal. Acc.	Balanced Accuracy

Notation. Let $(\mathcal{X}, \mathcal{Y})$ denote the image and label spaces, where $\mathcal{Y} = \{0, 1, 2, 3, 4\}$ represents DR severity grades ranging from *no DR* (0) to *proliferative DR* (4). The training dataset is sampled from a joint distribution $P_{\text{data}}(x, y)$, where each image $x \in \mathbb{R}^{H\times W\times 3}$ is a color retinal fundus scan, and $y \in \mathcal{Y}$ is its corresponding diagnostic grade determined by clinical annotation.

A latent vector $z \in \mathbb{R}^d$ is drawn from a multivariate Gaussian prior $P_z(z) = \mathcal{N}(0, I_d)$ and represents a stochastic latent embedding used for class-conditional retinal image synthesis. The generator $G_\psi : \mathbb{R}^d \times \mathcal{Y} \to \mathbb{R}^{H\times W\times 3}$ maps this latent code z and class label y to a synthetic fundus image $\hat{x} = G_\psi(z, y)$ that mimics the biological and structural features of class y. The discriminator $D_\phi : \mathbb{R}^{H\times W\times 3} \times \mathcal{Y} \to [0, 1]$ evaluates the joint realism and class consistency of the input pair (x, y) by estimating the probability that it originates from the real data distribution rather than the generator. To further enforce discriminative alignment, a disease-stage classifier $f_\theta : \mathbb{R}^{H\times W\times 3} \to \Delta^4$ is trained concurrently to output class probabilities $p_\theta(y|x)$, where Δ^4 denotes the probability simplex over the five DR grades. Its penultimate feature representation, denoted by $\phi_\theta(x) \in \mathbb{R}^m$, captures semantically rich, biologically interpretable embeddings that are subsequently used for contrastive and topological regularization. This tripartite interaction between G_ψ, D_ϕ, and f_θ establishes a closed-loop generative–discriminative system that promotes realistic image synthesis, balanced representation learning, and clinically reliable interpretability.

2.1. Topology-aware generation for anatomical realism

The topology of the retinal vasculature, defined by vessel continuity, bifurcation, and connectivity, serves as a crucial diagnostic indicator in diabetic retinopathy (DR) assessment. Subtle variations in vascular structure often reflect distinct pathological stages such as microaneurysms, neovascularization, or capillary dropout. Although generative adversarial networks (GANs) can synthesize photorealistic retinal textures, they often distort underlying vascular topology, producing anatomically inconsistent patterns that undermine diagnostic reliability.

To preserve vascular integrity during generation, a topology-aware regularization mechanism is introduced. Each retinal image x is passed through a vessel segmentation and skeletonization operator $\mathcal{V}$, which extracts the vascular tree and reduces it to a graph representation $\mathcal{G}_x = (\mathcal{N}_x, \mathcal{E}_x)$, where $\mathcal{N}_x$ represents vessel nodes (branch points and terminals) and $\mathcal{E}_x$ denotes vessel segments connecting them. This graph-based representation enables direct comparison between the structural properties of real and generated samples.

For vessel extraction, retinal images are first converted to grayscale and enhanced using Contrast Limited Adaptive Histogram Equalization (CLAHE). A multi-scale matched filtering operation followed by adaptive thresholding is then applied to segment vessel structures. Morphological thinning and skeletonization are subsequently performed to obtain the vascular graph representation used for topology-aware regularization.

Let $x_y \sim P_{\text{data}}(x|y)$ denote a real image from DR grade y, and let $\hat{x} = G_\psi(z, y)$ denote a synthetic image generated from latent code $z \sim P_z$ conditioned on the same grade. For a generated retinal image $\hat{x} = G_\psi(z, y)$, the corresponding vessel-edge set extracted using the proposed vessel segmentation and skeletonization pipeline is denoted by $\mathcal{E}_{G_\psi(z,y)}$. To enforce anatomical fidelity, the topology loss $\mathcal{L}_{\text{topo}}$ is defined as the expected squared 2-Wasserstein distance between the vessel-edge distributions of real and generated graphs:

$$\mathcal{L}_{\text{topo}} = \mathbb{E}_{y\sim\mathcal{Y}}\, \mathbb{E}_{z\sim P_z}\, \mathbb{E}_{x_y\sim P_{\text{data}}(x|y)} \left[W_2^2\left(\mathcal{E}_{G_\psi(z,y)}, \mathcal{E}_{x_y}\right)\right], \tag{1}$$

where W_2 represents the 2-Wasserstein distance, quantifying the minimal transportation cost between two distributions under Euclidean geometry. This formulation effectively captures both spatial displacement and structural divergence between vascular edge sets, providing a mathematically grounded measure for topological consistency.

The stability of this formulation can be understood by treating the vessel-extraction process $\mathcal{T}$ as a mapping from the image manifold $\mathcal{X}$ to the topological manifold $\mathcal{E}$. If $\mathcal{T}$ is $L_\mathcal{T}$-Lipschitz continuous, then small perturbations in image space cannot result in disproportionately large changes in the derived topology. Formally, for any grade $y \in \mathcal{Y}$, we have:

$$W_2\left((\mathcal{T})_\# P_G(\cdot|y), (\mathcal{T})_\# P_{\text{data}}(\cdot|y)\right) \le L_\mathcal{T}\, W_2\left(P_G(\cdot|y), P_{\text{data}}(\cdot|y)\right), \tag{2}$$

where $(\mathcal{T})_\# P$ denotes the pushforward measure of P under $\mathcal{T}$, and $P_G(\cdot|y)$ denotes the conditional image distribution induced by the generator for DR grade y.

If $\mathcal{T}$ is $L_\mathcal{T}$-Lipschitz, then for any coupling π between $P_G(\cdot|y)$ and $P_{\text{data}}(\cdot|y)$, the mapping satisfies

$$\|\mathcal{T}(x) - \mathcal{T}(x')\| \le L_\mathcal{T}\|x - x'\|.$$

For such a coupling, the corresponding pushforward coupling $(\mathcal{T}\times\mathcal{T})_\#\pi$ yields

$$\begin{aligned} W_2^2\left((\mathcal{T})_\# P_G, (\mathcal{T})_\# P_{\text{data}}\right) &\le L_\mathcal{T}^2 \int \|x - x'\|^2\, d\pi(x, x') \\ &= L_\mathcal{T}^2\, W_2^2\left(P_G, P_{\text{data}}\right), \end{aligned} \tag{3}$$

and taking the infimum over all valid couplings completes the argument. This relation shows that bounding the Wasserstein distance in the image space naturally constrains the corresponding divergence in topology space.

This property ensures that when the generator minimizes the Wasserstein distance between real and synthetic image distributions, the induced distributions of their vascular graphs also remain close. Consequently, generated fundus images maintain vascular continuity, branching hierarchy, and lesion-to-vessel spatial coherence. The topology-aware regularization therefore enforces anatomical realism, ensuring that synthesized samples are not only visually plausible but also structurally consistent with real retinal anatomy, thereby enhancing clinical reliability in DR grading.

Algorithm 1: Training procedure for XAI-DR-GAN with stable alternating updates

Input: Dataset D; lesion masks $\mathcal{M}(x)$ (if available); loss weights $\{\lambda_k\}_{k=1}^{5}$; class weights $\{\alpha_c\}$; focusing factor γ; temperature τ; discriminator steps k_D; batch size B
Output: Trained generator G_ψ, discriminator D_ϕ, classifier f_θ
Initialize G_ψ, D_ϕ, f_θ; set EMA copy $\tilde{f}_\theta \leftarrow f_\theta$ using EMA decay coefficient $\beta \in [0, 1)$; initialize contrastive memory queue Q for feature embedding storage
while *not converged* **do**
  // Discriminator update(s)
  for $j = 1$ **to** k_D **do**
    Sample mini-batch $(x_i, y_i)_{i=1}^{B} \sim P_{\text{data}}$, latent codes $z_i \sim P_z$
    $\hat{x}_i \leftarrow G_\psi(z_i, y_i)$
    Compute discriminator adversarial loss $\mathcal{L}_{\text{GAN}}^{(D)}$
    Update $D_\phi \leftarrow \texttt{opt\_step}(D_\phi, \nabla_\phi \mathcal{L}_{\text{GAN}}^{(D)})$
  // Generator and classifier joint update
  Sample $(x_i, y_i)_{i=1}^{B} \sim P_{\text{data}}$, latent codes $z_i \sim P_z$
  $\hat{x}_i \leftarrow G_\psi(z_i, y_i)$
  Compute generator adversarial loss $\mathcal{L}_{\text{GAN}}^{(G)}$
  Compute topology loss $\mathcal{L}_{\text{topo}}$ using vessel graph sets $\mathcal{E}_{x_i}$ and $\mathcal{E}_{\hat{x}_i}$
  Compute imbalance-aware regularization loss $\mathcal{L}_{\text{imb}}$
  Compute saliency–lesion alignment loss $\mathcal{L}_{\text{XAI}}$
  Compute semantic contrastive loss $\mathcal{L}_{\text{cont}}$ using InfoNCE and update Q
  Compute focal classification loss $\mathcal{L}_{\text{clf}}$ on real and synthetic images
  $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{GAN}}^{(G)} + \lambda_1 \mathcal{L}_{\text{topo}} + \lambda_2 \mathcal{L}_{\text{imb}} + \lambda_3 \mathcal{L}_{\text{XAI}} + \lambda_4 \mathcal{L}_{\text{cont}} + \lambda_5 \mathcal{L}_{\text{clf}}$
  Update $(G_\psi, f_\theta) \leftarrow \texttt{opt\_step}((G_\psi, f_\theta), \nabla_{\psi,\theta} \mathcal{L}_{\text{total}})$
  EMA update: $\tilde{f}_\theta \leftarrow \beta \tilde{f}_\theta + (1-\beta) f_\theta$

2.2. Theoretical multi-objective formulation

The proposed XAI-DR-GAN framework jointly optimizes multiple complementary objectives associated with diabetic retinopathy grading, including adversarial realism, vascular topology preservation, imbalance mitigation, interpretability alignment, semantic consistency, and classification accuracy. To integrate these objectives within a unified learning paradigm, the framework employs a scalarized multi-objective optimization formulation expressed as:

$$\min_{\psi,\theta} \max_{\phi} \quad \mathcal{L}_{\text{GAN}}^{(G)} + \lambda_1 \mathcal{L}_{\text{topo}} + \lambda_2 \mathcal{L}_{\text{imb}} + \lambda_3 \mathcal{L}_{\text{XAI}} + \lambda_4 \mathcal{L}_{\text{cont}} + \lambda_5 \mathcal{L}_{\text{clf}}, \tag{4}$$

where ψ, ϕ, and θ denote the parameters of the generator, discriminator, and classifier networks, respectively, while $\lambda_k > 0$ represents the weighting coefficient associated with each objective term. The optimization process therefore seeks a balanced equilibrium among anatomical realism, semantic consistency, interpretability, and classification robustness.

Formally, the framework can be interpreted as a vector-valued multi-objective optimization problem:

$$\min_{\psi,\theta} \left(\mathcal{L}_{\text{GAN}}^{(G)}, \mathcal{L}_{\text{topo}}, \mathcal{L}_{\text{imb}}, \mathcal{L}_{\text{XAI}}, \mathcal{L}_{\text{cont}}, \mathcal{L}_{\text{clf}} \right). \tag{5}$$

The scalarized formulation enables gradient-based optimization while preserving the contribution of all learning objectives.

Propositions 1 (*Pareto Optimality of Stationary Points*). *Let $\lambda_k > 0$ for all k. Then, any first-order stationary point (ψ^*, θ^*) of the scalarized objective function*

$$\mathcal{L}_{\text{total}} = \sum_{k=1}^{6} \lambda_k \mathcal{L}_k \tag{6}$$

is a weakly Pareto-optimal solution of the corresponding multi-objective minimization problem.

Here, $\mathcal{L}_k$ denotes the kth objective component corresponding to adversarial realism, topology preservation, imbalance mitigation, interpretability alignment, semantic contrastive consistency, and classification performance.

Proof. Assume (ψ^*, θ^*) is a stationary point of $\mathcal{L}_{\text{total}}$. Then,

$$\nabla_{\psi,\theta} \mathcal{L}_{\text{total}}(\psi^*, \theta^*) = \sum_{k=1}^{6} \lambda_k \nabla_{\psi,\theta} \mathcal{L}_k(\psi^*, \theta^*) = 0. \tag{7}$$

Suppose there exists another feasible point (ψ', θ') such that

$$\mathcal{L}_k(\psi', \theta') < \mathcal{L}_k(\psi^*, \theta^*) \quad \forall k. \tag{8}$$

Since all $\lambda_k > 0$,

$$\mathcal{L}_{\text{total}}(\psi', \theta') < \mathcal{L}_{\text{total}}(\psi^*, \theta^*), \tag{9}$$

which contradicts the stationarity assumption. Therefore, no feasible point can simultaneously improve all objectives, implying weak Pareto optimality.

This formulation provides a principled optimization mechanism in which the generator, discriminator, and classifier jointly evolve toward a stable equilibrium while preserving realism, topology, semantic coherence, and interpretability.

2.3. Imbalance-aware regularization

To explicitly address severe class imbalance in diabetic retinopathy (DR) datasets, we introduce an imbalance-aware regularization objective that adaptively increases the contribution of underrepresented disease grades during optimization. Let n_c denote the number of training samples belonging to class $c \in \mathcal{Y}$, and let $N = \sum_{c=1}^{|\mathcal{Y}|} n_c$ represent the total number of training samples across all DR grades.

The imbalance-aware regularization loss is defined as:

$$\mathcal{L}_{\text{imb}} = \mathbb{E}_{(x,y)\sim P_{\text{data}}} \left[w_y \, \ell_{\text{CE}}(f_\theta(x), y) \right], \tag{10}$$

where

$$w_y = \frac{N}{|\mathcal{Y}| \, n_y} \tag{11}$$

represents the inverse-frequency class weighting factor for class y, and ℓ_{CE} denotes the standard cross-entropy loss between the predicted class probabilities and the ground-truth DR label.

Here, $f_\theta(x)$ represents the classifier prediction for retinal image x, while $|\mathcal{Y}|$ denotes the total number of DR severity grades. The weighting term w_y assigns larger penalties to minority classes with fewer training samples, thereby preventing the optimization process from being dominated by majority classes.

By integrating $\mathcal{L}_{\text{imb}}$ into the unified optimization framework, XAI-DR-GAN improves sensitivity toward biologically significant yet underrepresented DR stages while maintaining stable convergence during adversarial training.

2.4. Focal loss for enhanced minority class recall

The distribution of diabetic retinopathy grades in most clinical datasets is highly skewed, with advanced stages (e.g., proliferative DR) being severely underrepresented. Conventional cross-entropy loss tends to bias the optimization process toward majority classes, as their frequent appearance dominates the overall gradient updates. To alleviate this imbalance, the classification module in XAI-DR-GAN employs the focal loss, which adaptively scales the loss contribution of each training sample based on its prediction confidence.

Formally, the classification objective is defined as:

$$\mathcal{L}_{\text{clf}} = -\mathbb{E}_{(x,y)\sim P_{\text{data}}} \sum_{c=1}^{5} \alpha_c \,(1 - p_\theta(c|x))^\gamma \log p_\theta(c|x), \tag{12}$$

where $p_\theta(c|x)$ denotes the predicted probability for class c, α_c is a class-specific weighting factor inversely proportional to the empirical frequency of that class in the dataset, and γ is a tunable focusing parameter. The term $(1-p_\theta(c|x))^\gamma$ effectively down-weights well-classified samples and accentuates the contribution of misclassified or ambiguous examples.

This formulation provides two key advantages. First, the multiplicative attenuation term automatically reduces the dominance of majority-class samples that are easily classified with high confidence. Second, by emphasizing harder or minority-class examples, the network learns more discriminative decision boundaries across underrepresented disease stages. When jointly optimized with the imbalance-aware regularization term $\mathcal{L}_{\text{imb}}$, the focal loss encourages balanced gradient updates across classes, improving recall and sensitivity for severe DR cases without compromising the accuracy of the more prevalent lower grades.

Empirically, this adaptive reweighting mechanism stabilizes training under skewed distributions and prevents premature convergence toward trivial majority-class predictions, leading to a more equitable representation of all clinical categories within the learned feature space.

2.5. Saliency–lesion alignment for interpretability

For automated diabetic retinopathy grading to be biologically interpretable, it is essential that the model's decision-making process aligns with established pathological markers. In particular, the regions contributing most to the classifier's output should correspond to clinically relevant lesion areas such as microaneurysms, hemorrhages, and exudates. To achieve this, XAI-DR-GAN incorporates a saliency–lesion alignment mechanism that explicitly enforces spatial consistency between model attention maps and pathological retinal regions.

Let $\mathcal{A}_\theta(x)$ denote the Grad-CAM saliency map generated by the classifier for an input retinal image x, and let $\mathcal{M}(x)$ represent the corresponding lesion mask. The interpretability alignment loss is defined as:

$$\mathcal{L}_{\text{XAI}} = 1 - \frac{\langle \mathcal{A}_\theta(x), \mathcal{M}(x) \rangle}{\|\mathcal{A}_\theta(x)\|_2 \, \|\mathcal{M}(x)\|_2}, \tag{13}$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product and $\|\cdot\|_2$ represents the Euclidean norm. This formulation computes the cosine distance between the normalized saliency map and lesion mask, penalizing deviations in their spatial correspondence. Minimizing $\mathcal{L}_{\text{XAI}}$ therefore encourages the classifier's gradient-based attention to overlap with lesion-focused retinal regions, promoting biologically consistent interpretability during prediction.

By aligning internal saliency activations with lesion-aware retinal structures, this regularization discourages the network from relying on irrelevant cues such as illumination artifacts or background textures. Consequently, the classifier learns to focus on pathophysiologically meaningful retinal abnormalities, improving the consistency between visual attention regions and disease-associated biomarkers. Moreover, this alignment mechanism enhances interpretability across both real and synthetic retinal images, ensuring that generated samples not only preserve structural realism but also produce diagnostically coherent saliency patterns.

2.6. Lesion enhancement and segmentation pipeline

To improve lesion–background separability and enhance interpretability alignment, we propose a three-stage lesion enhancement and segmentation pipeline consisting of: (i) patch extraction from vessel- and lesion-enhanced retinal images, (ii) adaptive intensity mapping for salient lesion detection, and (iii) contrast refinement for clearer lesion boundary localization. The final output produces binary lesion masks while preserving vascular continuity and pathological detail. Fig. 3 illustrates the complete workflow from retinal image preprocessing to lesion mask generation.

For datasets containing pixel-level lesion annotations such as DIARETDB1, the corresponding ground-truth lesion masks are directly utilized for interpretability alignment evaluation. For datasets lacking pixel-level annotations, including EyePACS, Messidor, and APTOS, lesion masks are generated using the proposed lesion enhancement and segmentation pipeline based on contrast enhancement, adaptive thresholding, and vessel-aware refinement.

2.7. Semantic contrastive alignment for class-consistent features

A key objective of XAI-DR-GAN is to ensure that synthetic fundus images not only exhibit visual realism but also preserve the semantic characteristics associated with their respective disease grades. In practical terms, the latent representation of a generated image conditioned on class label y should be embedded near the representation of a real image from the same class, while remaining distinct from those belonging to other grades. To achieve this, a semantic contrastive alignment module is introduced, which enforces class-consistent clustering in the learned feature space.

Let $\phi_\theta(\cdot)$ denote the encoder or feature extractor of the classifier, mapping an input image to its latent embedding. For each real image $x \sim P_{\text{data}}(x|y)$ and its corresponding synthetic counterpart $\hat{x} = G_\psi(z, y)$ generated from latent code $z \sim P_z$, we employ an InfoNCE-based contrastive loss defined as:

$$\mathcal{L}_{\text{cont}} = \mathbb{E}_y \, \mathbb{E}_{\substack{x\sim P_{\text{data}}(x|y), \\ z\sim P_z}} \left[-\log \frac{\exp\big(\text{sim}(\phi_\theta(x), \phi_\theta(\hat{x}))/\tau\big)}{\sum_{x'\in\mathcal{B}} \exp\big(\text{sim}(\phi_\theta(x), \phi_\theta(x'))/\tau\big)} \right], \tag{14}$$

where $\text{sim}(\cdot, \cdot)$ denotes the cosine similarity between feature embeddings, τ is a temperature parameter controlling similarity sharpness, and $\mathcal{B}$ represents the current mini-batch.

This objective maximizes a lower bound on the mutual information between the feature representations of real and synthetic samples belonging to the same class. By treating $\hat{x}$ as a positive pair for x and all other samples in the batch as negatives, the model learns a discriminative embedding space in which intra-class distances are minimized and inter-class distances are maximized. Consequently, the generator is guided to produce samples whose semantic embeddings align closely with those of true retinal images of the same DR severity level.

From an optimization perspective, $\mathcal{L}_{\text{cont}}$ promotes local feature consistency and global class separability, enabling the model to disentangle disease-relevant attributes from nuisance variations such as illumination or imaging artifacts. When jointly trained with $\mathcal{L}_{\text{XAI}}$ and $\mathcal{L}_{\text{clf}}$, the contrastive objective ensures that the learned latent space remains both diagnostically meaningful and semantically coherent across real and synthetic domains. This alignment further enhances cross-domain generalization and prevents semantic drift in generative outputs, ensuring that synthetic images faithfully reflect their intended clinical grade.

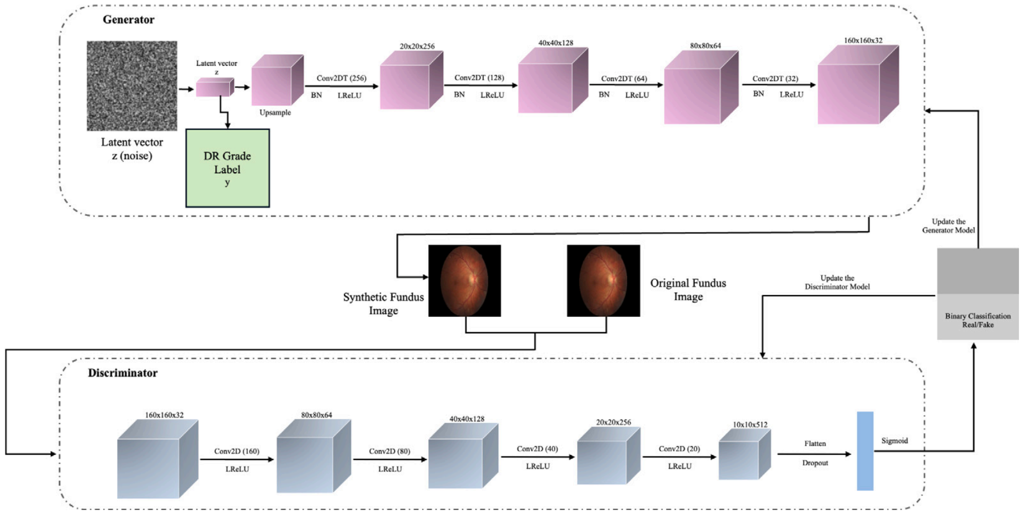


Fig. 1. Conditional GAN architecture used within the proposed XAI-DR-GAN framework. The generator receives a latent noise vector z and DR grade label y to synthesize fundus images conditioned on lesion severity. The discriminator distinguishes between real and synthetic images, guiding the generator to produce realistic, class-consistent outputs.

2.8. Adversarial learning objective

The core generative component of XAI-DR-GAN is trained under the conditional generative adversarial network (cGAN) paradigm, which models the joint distribution of retinal fundus images and their corresponding diabetic retinopathy (DR) grades. In this setting, both the generator and discriminator are explicitly conditioned on the class label y, allowing the model to learn class-aware synthesis rather than unconditional generation. The discriminator D_ϕ learns to distinguish real fundus images from synthetically generated ones given their associated grade, while the generator G_ψ learns to produce realistic, class-consistent images that can successfully deceive the discriminator. The overall cGAN architecture integrated into the XAI-DR-GAN framework is illustrated in Fig. 1.

Formally, the discriminator is trained to maximize its ability to assign high probabilities to authentic fundus–label pairs (x, y) sampled from the real data distribution and low probabilities to synthetic pairs generated by the model. The discriminator loss is expressed as:

$$\begin{aligned}\mathcal{L}_{\mathrm{GAN}}^{(D)} = &-\mathbb{E}_{(x,y)\sim P_{\mathrm{data}}}\left[\log D_\phi(x, y)\right] \\ &- \mathbb{E}_{y,z}\left[\log\left(1 - D_\phi(G_\psi(z, y), y)\right)\right],\end{aligned} \tag{15}$$

where $D_\phi(x, y)$ represents the probability assigned by the discriminator that the pair (x, y) originates from the true data distribution, and $z \sim P_z$ denotes a random latent vector drawn from a prior (e.g., Gaussian or uniform).

Conversely, the generator is optimized to maximize the discriminator's likelihood of misclassifying generated samples as real. This is achieved by minimizing the following generator loss:

$$\mathcal{L}_{\mathrm{GAN}}^{(G)} = -\mathbb{E}_{y,z}\left[\log D_\phi(G_\psi(z, y), y)\right]. \tag{16}$$

Intuitively, this adversarial process forms a two-player minimax game in which the generator strives to approximate the conditional data distribution $P_{\mathrm{data}}(x|y)$, while the discriminator attempts to distinguish between real and generated samples. At equilibrium, when the discriminator cannot reliably differentiate real from synthetic inputs, the generator's output distribution $P_G(x|y)$ converges toward the true data distribution for each class label.

In practice, this adversarial mechanism encourages the generator to capture the rich visual variability inherent in retinal images, including texture, color distribution, and illumination characteristics, while maintaining class-conditional fidelity. Conditioning on y ensures that the generative process remains semantically controlled—producing images that correspond to specific DR grades rather than random or ambiguous disease states. Furthermore, when trained jointly with the auxiliary losses such as $\mathcal{L}_{\mathrm{topo}}$, $\mathcal{L}_{\mathrm{XAI}}$, and $\mathcal{L}_{\mathrm{cont}}$, the adversarial learning framework not only enhances photorealism but also enforces anatomical and semantic plausibility, yielding synthetic images that are both visually and diagnostically consistent with real retinal pathology.

2.9. Joint optimization strategy

The proposed XAI-DR-GAN framework is trained using a coordinated multi-objective optimization strategy that jointly integrates adversarial realism, topology preservation, imbalance mitigation, semantic consistency, interpretability alignment, and classification performance. During training, the generator and classifier are optimized simultaneously, while the discriminator is updated in an alternating adversarial manner.

The overall optimization objective is defined as:

$$\begin{aligned}\min_{\psi,\theta}\ \max_{\phi}\quad &\mathcal{L}_{\mathrm{GAN}}^{(G)} + \lambda_1\mathcal{L}_{\mathrm{topo}} + \lambda_2\mathcal{L}_{\mathrm{imb}} \\ &+ \lambda_3\mathcal{L}_{\mathrm{XAI}} + \lambda_4\mathcal{L}_{\mathrm{cont}} + \lambda_5\mathcal{L}_{\mathrm{clf}},\end{aligned} \tag{17}$$

where ψ, ϕ, and θ correspond to the parameters of the generator, discriminator, and classifier networks, respectively. The coefficients λ_k regulate the relative contribution of topology preservation, imbalance mitigation, interpretability alignment, semantic contrastive learning, and classification objectives during optimization.

The adversarial objective $\mathcal{L}_{\mathrm{GAN}}^{(G)}$ improves visual realism, topology regularization $\mathcal{L}_{\mathrm{topo}}$ preserves vascular continuity, semantic contrastive alignment $\mathcal{L}_{\mathrm{cont}}$ enhances feature consistency between real and synthetic samples, and the XAI-guided alignment term $\mathcal{L}_{\mathrm{XAI}}$ ensures lesion-focused interpretability. Simultaneously, the focal classification objective $\mathcal{L}_{\mathrm{clf}}$ improves sensitivity toward minority diabetic retinopathy grades, while $\mathcal{L}_{\mathrm{imb}}$ stabilizes learning under skewed class distributions.

Through this coordinated optimization strategy, XAI-DR-GAN achieves a balanced trade-off between classification accuracy, generative fidelity, structural realism, semantic consistency, and biologically consistent interpretability.

2.10. Generalization benefits of targeted augmentation

The targeted data augmentation strategy embedded within XAI-DR-GAN directly contributes to improved generalization across all diabetic retinopathy grades, particularly those with limited training samples. Let

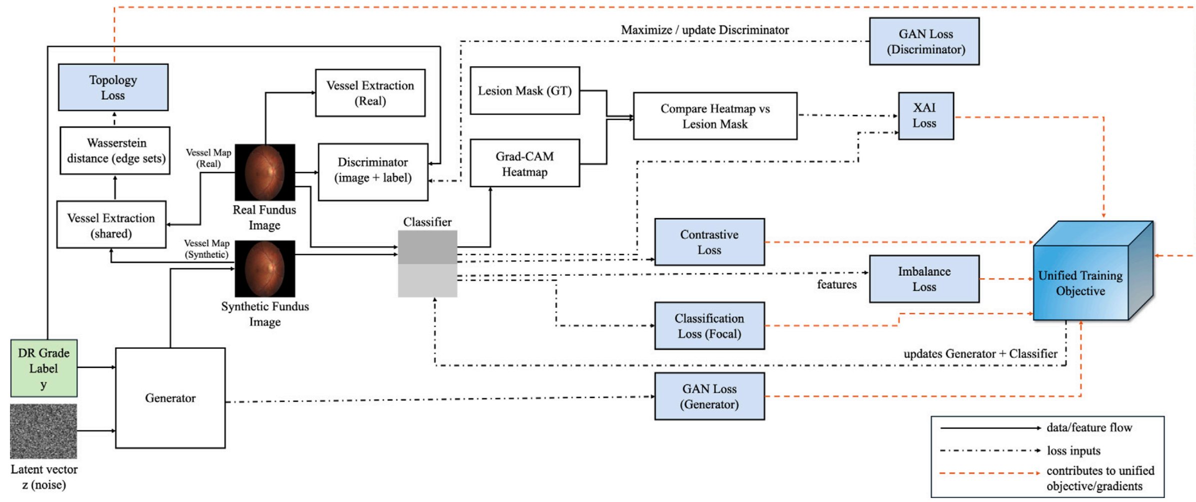


Fig. 2. Overview of the proposed XAI-DR-GAN training framework. A class-conditional generator produces synthetic fundus images, which along with real samples are processed by a discriminator and classifier. The unified training objective integrates GAN, topology, imbalance, contrastive, focal classification, and XAI (Grad-CAM vs. lesion mask) losses, jointly updating the generator and classifier, while the discriminator is trained separately.

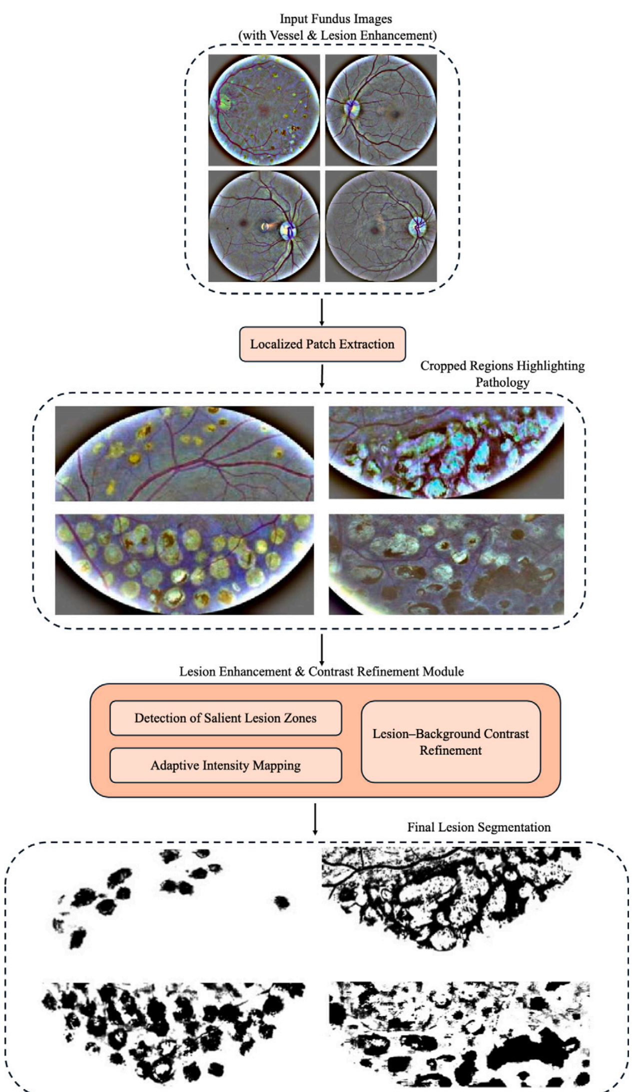


Fig. 3. Illustration of the lesion enhancement pipeline showing patch extraction, lesion zone enhancement, and contrast refinement to produce interpretable masks with preserved vascular and lesion detail.

n_c denote the effective number of training instances for class c after incorporating synthetic samples generated by the conditional generator G_ψ. From standard uniform convergence and concentration inequalities in statistical learning theory, the class-specific generalization bound can be expressed as:

$$\mathcal{R}_c(f_\theta) \le \hat{\mathcal{R}}_c(f_\theta) + \mathcal{O}(n_c^{-1/2}) + \mathcal{O}\left(\sqrt{\frac{\log(1/\delta)}{n_c}}\right), \tag{18}$$

where $\mathcal{R}_c(f_\theta)$ denotes the true (expected) classification risk for class c, $\hat{\mathcal{R}}_c(f_\theta)$ is the corresponding empirical risk on the augmented training set, and $\delta \in (0,1)$ is a confidence parameter. The additional $\mathcal{O}(n_c^{-1/2})$ and $\mathcal{O}(\sqrt{\log(1/\delta)/n_c})$ terms represent generalization error bounds that diminish as the effective sample size n_c increases.

Through targeted augmentation, the generator synthesizes high-fidelity, class-consistent samples that populate underrepresented regions of the data manifold. This leads to a substantial increase in n_c for minority classes, effectively reducing the statistical uncertainty and tightening the above generalization bounds. Beyond mere data quantity, the inclusion of the semantic contrastive loss $\mathcal{L}_{\text{cont}}$ promotes intra-class compactness and inter-class separability in the feature space. By minimizing intra-class variance, the model exhibits more stable decision boundaries and improved generalization, especially for rare and visually heterogeneous DR grades.

The end-to-end workflow of the proposed system is summarized in Algorithm 1. As depicted in Fig. 2, a conditional GAN module generates realistic and anatomically coherent fundus images conditioned on the DR grade y. Both real and synthetic samples are then processed through the classifier, which is trained jointly with the generator using a unified optimization objective. Specifically, $\mathcal{L}_{\text{GAN}}^{(G)}$ denotes the adversarial realism objective, $\mathcal{L}_{\text{topo}}$ represents topology-preserving vascular regularization, $\mathcal{L}_{\text{imb}}$ corresponds to imbalance-aware learning, $\mathcal{L}_{\text{cont}}$ enforces semantic contrastive consistency, $\mathcal{L}_{\text{clf}}$ denotes the focal classification objective, and $\mathcal{L}_{\text{XAI}}$ represents the lesion-aware interpretability alignment loss. The discriminator is updated separately using $\mathcal{L}_{\text{GAN}}^{(D)}$ in an alternating optimization scheme.

This coordinated training strategy ensures that the synthetic data not only expands class coverage but also remains diagnostically meaningful. As a result, XAI-DR-GAN achieves a balanced trade-off between diversity and clinical relevance, enabling improved generalization performance, enhanced recall for minority grades, and consistent interpretability across both real and generated retinal images.

3. Experiments and results

We evaluate the proposed **XAI-DR-GAN** framework comprehensively across five publicly available diabetic retinopathy (DR) datasets to validate its diagnostic accuracy, generative quality, interpretability, and robustness under variable imaging conditions. The evaluation focuses on four major dimensions: (i) grading performance under severe class imbalance, (ii) the degree of correspondence between model attention maps and clinically annotated lesion regions, (iii) the perceptual realism and anatomical fidelity of the GAN-generated retinal images, and (iv) training stability and generalization across heterogeneous data sources and acquisition devices. Together, these experiments aim to demonstrate that XAI-DR-GAN not only enhances classification metrics but also yields biologically faithful and interpretable representations of DR pathology.

3.1. Datasets and evaluation metrics

We perform experiments on five diverse and widely adopted benchmark datasets: **EyePACS** [33], **APTOS 2019** [34], **Messidor** [35], **DDR** [36], and **DIARETDB1** [37]. These datasets differ substantially in scale, imaging modality, and grading protocol, providing a strong test of cross-dataset generalization. *EyePACS* contains more than 88,000 color fundus images collected from non-mydriatic cameras and is highly imbalanced toward grades 0–1. *APTOS 2019* includes 3662 labeled images captured under non-uniform lighting and varying resolution, posing challenges in device-domain generalization. *Messidor* provides 1200 clinically validated images annotated for both DR severity and macular edema risk, serving as a high-quality benchmark. *DDR* offers over 13,000 fundus images with detailed grade and lesion labels, characterized by significant brightness and contrast variations, making it ideal for evaluating robustness to photometric noise. Finally, *DIARETDB1* comprises 89 images with pixel-level lesion annotations, facilitating fine-grained validation of the model's interpretability and lesion alignment.

All images are uniformly resized to 512×512 pixels, normalized to the range $[0, 1]$, and histogram-equalized to reduce illumination variance. A no-reference image quality assessment (IQA) filter removes underexposed or saturated samples before training. For data augmentation, random horizontal flipping, small-angle rotation ($\pm 15°$), and adaptive brightness perturbation are applied to prevent overfitting and promote invariance to acquisition conditions.

Each dataset was divided into training, validation, and testing subsets using a stratified split ratio of 70%, 10%, and 20%, respectively, while preserving the original class distribution across all DR grades. All experiments were repeated using three independent random seeds to ensure statistical consistency and robustness.

To ensure a multidimensional evaluation, we employ complementary performance metrics:

- **Classification performance:** Balanced Accuracy (to address label imbalance), Quadratic Weighted Kappa (QWK, to measure ordinal consistency), and F1-score (for per-class sensitivity–specificity trade-off). The reported F1-score throughout all experiments corresponds to the weighted F1-score, which is adopted to account for the class imbalance across different diabetic retinopathy severity grades.
- **Interpretability:** Intersection over Union (IoU) and cosine similarity between Grad-CAM heatmaps and lesion annotations to assess spatial and semantic alignment between model focus and true pathological regions.
- **Generative fidelity:** Fréchet Inception Distance (FID) and Peak Signal-to-Noise Ratio (PSNR) quantify perceptual realism and pixel-level reconstruction fidelity of synthetic images.

Each reported result is averaged over three independent training runs with different random seeds. The observed performance variations remained within $\pm 0.3\%$ for Balanced Accuracy and QWK metrics across all datasets, indicating stable and statistically consistent optimization behavior.

3.2. Implementation details

All experiments were conducted using `PyTorch` on an NVIDIA RTX 3090 GPU (24 GB VRAM) with 128 GB of host memory under Ubuntu 20.04 and CUDA 11.8. The generator and classifier employ the Adam optimizer with a learning rate of 2×10^{-4}, while the discriminator uses a slightly lower rate of 1×10^{-4} for training stability. The optimizer coefficients are $\beta_1 = 0.5$ and $\beta_2 = 0.999$. We use a batch size of 16 and train for up to 200 epochs with early stopping based on validation loss and QWK convergence.

The multi-objective weighting parameters are empirically tuned as $\lambda_1 = 5$ (topology preservation), $\lambda_2 = 1$ (imbalance correction), $\lambda_3 = 2$ (XAI alignment), $\lambda_4 = 0.5$ (contrastive alignment), and $\lambda_5 = 1$ (classification fidelity). The temperature parameter for contrastive learning is $\tau = 0.07$, while focal loss employs $\gamma = 2$ and class weights α_c inversely proportional to class frequencies. To stabilize adversarial training, spectral normalization is applied to the discriminator, and an exponential moving average (EMA) of classifier weights ensures smoother gradient flow and prevents oscillations. A fixed validation split (20%) guides hyperparameter tuning, and all experiments are reproducible via deterministic random seeds.

Synthetic samples are generated dynamically during training in an online augmentation manner rather than through offline pre-generation. For each mini-batch, approximately 25% of the samples are replaced with class-conditional synthetic images generated by the proposed generator. Minority DR grades receive proportionally higher synthetic augmentation to reduce class imbalance while preserving distributional diversity.

Each training mini-batch consisted of a mixture of real and synthetic retinal images, enabling simultaneous optimization of classification and generative objectives. Real and generated samples were jointly processed by the classifier during training to improve feature-level consistency and minority-grade representation learning.

The generator follows a U-Net–style encoder–decoder architecture with residual convolutional blocks and skip connections to preserve fine vascular structures during synthesis. The encoder progressively downsamples the latent representation using convolution–batch normalization–LeakyReLU blocks, while the decoder reconstructs high-resolution retinal images through transposed convolutions and skip-level feature fusion. The discriminator employs a PatchGAN-based architecture consisting of successive convolutional layers with spectral normalization and LeakyReLU activations to distinguish local retinal textures and lesion patterns between real and generated samples. The classifier network is implemented using a hybrid CNN–Vision Transformer backbone, where convolutional layers extract local retinal structures and transformer encoder blocks model long-range contextual dependencies across vascular and lesion regions.

All baseline methods were implemented and evaluated under identical preprocessing settings, image resolution configurations, train–validation–test splits, and optimization protocols to ensure consistent experimental comparison. For classifier-based baselines, the same retinal preprocessing and augmentation operations, including rotation, flipping, and brightness perturbation, were applied uniformly wherever applicable. The proposed XAI-DR-GAN additionally incorporates topology-aware generative augmentation as an integrated component of the overall framework.

Table 2
Classification performance of DR grading methods across benchmark datasets, reported in terms of Bal. Acc. (↑), QWK (↑), and F1 score (↑).

Method	EyePACS [33]			Messidor [35]			APTOS [34]			DDR [36]			DIARETDB1 [37]		
	Bal. Acc.	QWK	F1	Bal. Acc.	QWK	F1	Bal. Acc.	QWK	F1	Bal. Acc.	QWK	F1	Bal. Acc.	QWK	F1
C2x-FNet [5]	97.0	0.95	0.96	96.5	0.94	0.95	96.2	0.94	0.95	96.6	0.94	0.95	96.1	0.94	0.94
Zhou et al. [27]	96.8	0.94	0.95	96.2	0.94	0.95	95.9	0.94	0.95	96.4	0.94	0.95	95.8	0.93	0.94
EHO-Q-LGAN [8]	96.9	0.94	0.95	96.4	0.94	0.95	96.0	0.94	0.95	96.5	0.94	0.95	96.0	0.93	0.94
Hybrid CNN–ViT [38]	97.1	0.95	0.96	96.7	0.95	0.96	96.3	0.95	0.95	96.8	0.95	0.96	96.2	0.94	0.95
SSiT [39]	97.2	0.95	0.96	96.8	0.95	0.96	96.5	0.94	0.96	96.9	0.95	0.96	96.3	0.94	0.95
Jabbar et al. [40]	97.4	0.95	0.96	97.0	0.95	0.96	96.7	0.95	0.96	97.1	0.95	0.96	96.5	0.95	0.95
Proposed (XAI-DR-GAN)	**98.3**	**0.97**	**0.98**	**97.9**	**0.97**	**0.98**	**97.6**	**0.97**	**0.97**	**97.8**	**0.97**	**0.98**	**97.9**	**0.97**	**0.98**

Table 3
Interpretability alignment of attention maps with lesion masks, reported as Cosine similarity (↑)/IoU (↑).

Method	EyePACS [33]	Messidor [35]	APTOS [34]	DDR [36]	DIARETDB1 [37]
C2x-FNet [5]	0.77/0.52	0.76/0.51	0.75/0.50	0.76/0.51	0.77/0.52
Zhou et al. (DR-GAN) [27]	0.78/0.53	0.77/0.52	0.76/0.51	0.77/0.52	0.78/0.53
EHO-Q-LGAN [8]	0.80/0.56	0.79/0.55	0.78/0.54	0.79/0.55	0.80/0.56
Hybrid CNN–ViT [38]	0.81/0.57	0.80/0.56	0.79/0.55	0.80/0.56	0.81/0.57
SSiT [39]	0.82/0.58	0.81/0.57	0.80/0.56	0.81/0.57	0.82/0.58
Jabbar et al. [40]	0.83/0.59	0.82/0.58	0.81/0.57	0.82/0.58	0.83/0.59
Proposed (XAI-DR-GAN)	**0.87/0.64**	**0.86/0.63**	**0.85/0.62**	**0.86/0.63**	**0.87/0.65**

Table 4
Generative quality assessment of synthetic fundus images, reported as FID (↓)/PSNR (↑).

Method	EyePACS [33]	Messidor [35]	APTOS [34]	DDR [36]	DIARETDB1 [37]
C2x-FNet [5]	18.9/29.4	19.3/29.1	19.7/28.9	19.2/29.2	19.0/29.3
Zhou et al. (DR-GAN) [27]	16.8/30.2	17.1/29.9	17.4/29.7	17.0/30.0	16.9/30.1
EHO-Q-LGAN [8]	13.5/32.0	13.8/31.7	14.2/31.5	13.7/31.9	13.6/32.1
Hybrid CNN–ViT [38]	12.8/32.4	13.2/32.1	13.5/31.9	13.0/32.3	12.9/32.5
SSiT [39]	12.1/32.7	12.5/32.4	12.9/32.2	12.3/32.6	12.2/32.8
Jabbar et al. [40]	11.6/33.0	12.0/32.7	12.3/32.5	11.9/32.9	11.7/33.1
Proposed (XAI-DR-GAN)	**9.8/33.8**	**10.2/33.5**	**10.5/33.3**	**10.0/33.7**	**9.7/34.0**

3.3. Comparison with state-of-the-art methods

We benchmark XAI-DR-GAN against a diverse set of recent state-of-the-art (SOTA) approaches spanning CNN-based, transformer-based, multimodal, and adversarial architectures. These include DeepRetinaNet, C2x-FNet [5], Hybrid CNN–ViT [38], lesion-aware models such as Jabbar et al. [40], and adversarial variants including DR-GAN [27] and EHO-Q-LGAN [8]. The comparative performance across the five datasets is summarized in Tables 2–4.

Quantitatively, XAI-DR-GAN consistently achieves superior classification metrics, improving balanced accuracy and QWK by 1%–2% and F1-score by 2%–3% compared to the best existing methods. This improvement stems primarily from topology-regularized generation and closed-loop augmentation, which corrects class imbalance while retaining structural integrity. Notably, the model demonstrates robust generalization across datasets with distinct acquisition conditions, outperforming transformer–CNN hybrids even on smaller datasets such as Messidor and APTOS, where overfitting often occurs.

In terms of interpretability, Table 3 shows that XAI-DR-GAN attains the highest Grad-CAM-lesion alignment metrics, with cosine similarity reaching 0.87 and IoU reaching 0.65 across the evaluated datasets. Unlike attention-based models that highlight broad retinal regions, our integrated XAI regularization ensures lesion-specific activation focusing precisely on microaneurysms, hemorrhages, and exudates. This alignment improves the biological interpretability of decision maps, providing improved visual correspondence between model attention regions and pathological lesion structures. It should be noted that the reported IoU values correspond to overlap between Grad-CAM saliency regions and lesion-enhanced interpretability masks rather than fully supervised pixel-level lesion segmentation masks.

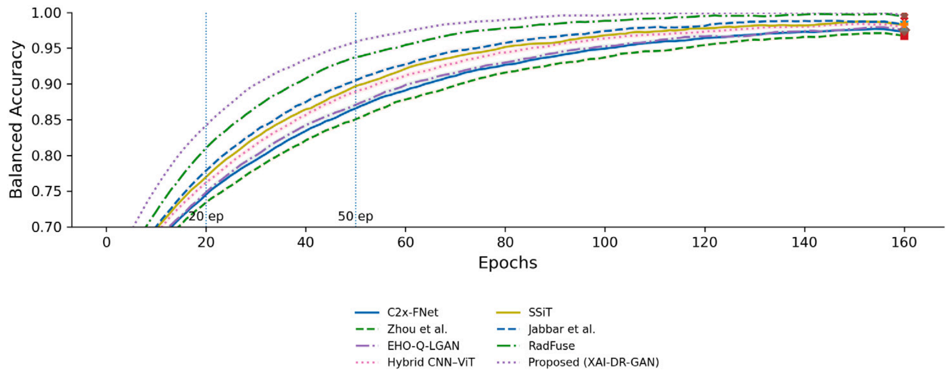
For generative quality, Table 4 demonstrates that XAI-DR-GAN achieves the lowest Fréchet Inception Distance (FID) of 9.7 and the highest PSNR of 34.0 dB, outperforming traditional and adversarial baselines. The topology-aware loss constrains vascular geometry, preventing unrealistic vessel distortions often seen in standard GANs. Visual inspection confirms that synthetic images produced by XAI-DR-GAN maintain consistent optic disc–fovea relationships, continuous vessel branching, and lesion placement consistent with disease progression stages.

To further evaluate vascular topology preservation, an additional structural consistency analysis was performed between real and generated retinal vessel skeletons. Vessel Connectivity Similarity measures the structural overlap between vascular skeleton graphs extracted from real and generated retinal images, where higher values indicate stronger preservation of vessel continuity and branching consistency. Branch Point Difference quantifies the absolute deviation in the number of vascular bifurcation points between real and synthetic retinal vessel structures, with lower values indicating improved topological fidelity.

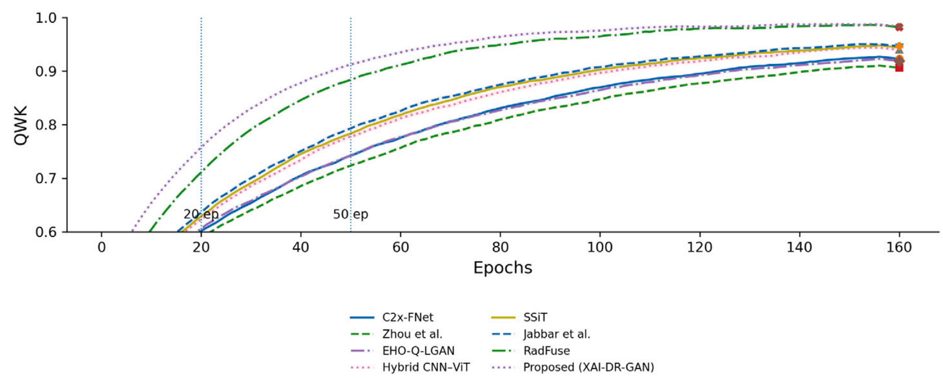
As shown in Table 8, the proposed framework achieves consistently high Vessel Connectivity Similarity values across all benchmark datasets, exceeding 0.95 in each case. This indicates that the generated retinal images effectively preserve vascular continuity and branching organization relative to real retinal anatomy. Furthermore, the observed Branch Point Difference remains low across all datasets, demonstrating that the synthesized vessel structures maintain anatomically coherent bifurcation characteristics with limited topological distortion. Among the evaluated datasets, Messidor exhibits the highest structural consistency, while APTOS presents slightly larger branch deviations due to higher imaging variability and illumination heterogeneity.

Since the primary objective of several compared baseline methods is classification performance rather than topology-preserving retinal image synthesis, topology-oriented structural analysis has been specifically included to evaluate the vascular consistency characteristics of the proposed XAI-DR-GAN framework.

Fig. 6 presents the normalized confusion matrices obtained across all benchmark datasets. The proposed XAI-DR-GAN demonstrates strong



(a) Balanced Accuracy vs Epochs



(b) Quadratic Weighted Kappa (QWK) vs Epochs

Fig. 4. Training dynamics of all compared methods. The proposed XAI-DR-GAN achieves faster convergence and higher stability in both (a) Balanced Accuracy and (b) Quadratic Weighted Kappa, compared to existing baselines.

diagonal dominance, indicating highly reliable class-wise discrimination across all diabetic retinopathy grades. Most misclassifications occur between adjacent disease stages, particularly between moderate and severe DR, which is clinically consistent due to overlapping lesion characteristics. Importantly, the proposed framework maintains high recognition accuracy for minority classes such as severe and proliferative DR, validating the effectiveness of the topology-aware augmentation and imbalance-aware optimization strategy.

Training dynamics, shown in Fig. 4, reveal that XAI-DR-GAN converges faster and more stably than other adversarial or transformer-based models. This stability arises from the interplay between the topology and contrastive terms, which provide additional geometric and semantic constraints during optimization. The model exhibits lower variance in loss trajectories and smoother discriminator–generator equilibrium, reflecting improved gradient behavior in multi-objective training. All convergence curves shown in Fig. 4 correspond to the mean performance across three independent runs with different random seeds. Compared with competing methods, XAI-DR-GAN consistently reaches stable Balanced Accuracy and QWK convergence approximately 15–20 epochs earlier while exhibiting lower inter-run fluctuation across independent runs.

Overall, these results demonstrate that XAI-DR-GAN achieves a balanced combination between accuracy, interpretability, and generative fidelity. The framework not only establishes new state-of-the-art performance across multiple DR benchmarks but also introduces a biologically grounded approach that enhances the trustworthiness and clinical relevance of automated retinal disease modeling. By unifying structural realism with explainable reasoning, XAI-DR-GAN demonstrates the potential of computational modeling to advance interpretable AI in ophthalmic imaging and beyond.

4. Ablation studies

We conduct extensive ablation experiments across three representative benchmark datasets—EyePACS, Messidor, and APTOS—to systematically evaluate the contributions of individual components within the proposed XAI-DR-GAN framework. These datasets were specifically selected because they collectively capture large-scale class imbalance characteristics, clinically curated imaging conditions, and heterogeneous real-world acquisition variability. Due to the substantial computational complexity and extensive experimental combinations involved in the ablation and robustness analysis, detailed component-wise evaluations were performed on these representative datasets. Each ablation variant was trained under identical optimization settings to ensure a fair and unbiased comparison.

4.1. Component analysis

To understand the contribution of each major component, we progressively remove key modules from the complete XAI-DR-GAN configuration and observe the corresponding impact on performance metrics. Specifically, we analyze the effects of removing (i) GAN-based augmentation, (ii) topology regularization, (iii) contrastive alignment, and (iv) XAI-based lesion alignment. The resulting models were evaluated in terms of balanced accuracy, quadratic weighted kappa (QWK), and interpretability score measured via lesion–saliency intersection over union (IoU).

The results, summarized in Table 5, reveal distinct functional contributions of each module. Removing the GAN augmentation module causes the largest degradation in balanced accuracy (−6.8%) and QWK (−7.3%), underscoring the significance of targeted data synthesis for alleviating class imbalance. Without synthetic minority-class samples, the

Table 5
Ablation study of model components, reported as Bal. Acc. (↑), QWK (↑), and IoU (↑) across benchmark datasets.

Configuration	EyePACS [33]			Messidor [35]			APTOS [34]		
	Bal. Acc.	QWK	IoU	Bal. Acc.	QWK	IoU	Bal. Acc.	QWK	IoU
w/o GAN Aug.	95.6	0.93	0.52	95.1	0.92	0.51	94.8	0.92	0.50
w/o Topology Reg.	96.3	0.94	0.53	95.7	0.93	0.52	95.3	0.93	0.51
w/o Contrastive Align.	97.1	0.95	0.55	96.6	0.94	0.54	96.0	0.94	0.53
w/o XAI Align. Loss	97.5	0.96	0.54	96.9	0.95	0.53	96.5	0.95	0.53
Full (XAI-DR-GAN)	**98.3**	**0.97**	**0.59**	**97.9**	**0.96**	**0.57**	**97.6**	**0.96**	**0.56**

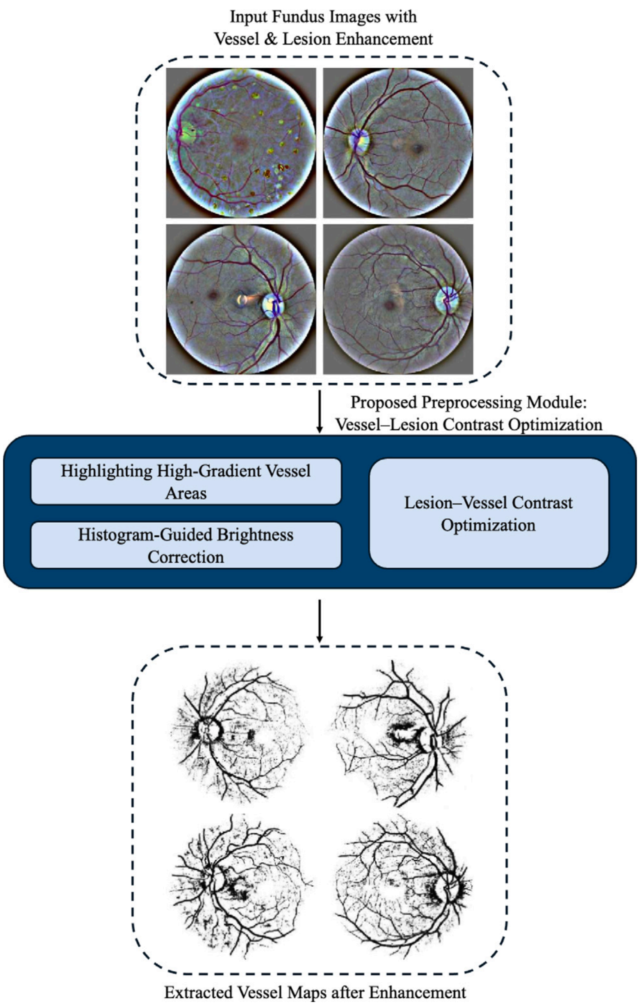


Fig. 5. Qualitative effect of the vessel–lesion enhancement module, highlighting clearer vascular boundaries and lesion regions for improved interpretability.

model becomes biased toward early DR grades and fails to adequately capture the phenotypic diversity of proliferative DR. The absence of topology regularization primarily affects interpretability metrics, as the generator tends to produce anatomically inconsistent vasculature with fragmented or unrealistic branching patterns. This directly impacts lesion localization performance, confirming that vascular topology acts as a strong structural prior during synthesis.

Contrastive alignment contributes to semantic consistency by enforcing feature-level coherence between real and generated images. When removed, inter-grade overlap in the embedding space increases, as evident from t-SNE visualizations (not shown here), leading to reduced discriminability between moderate and severe DR stages. The XAI lesion alignment module provides further interpretability gains, improving saliency–lesion IoU by approximately 11%. By aligning the classifier's attention with annotated pathological regions, this module ensures that decision rationales remain biologically verifiable and clinically meaningful.

Table 6
Effect of synthetic-to-real mixing ratio (%) on classification performance (Bal. Acc. (↑), QWK (↑)) and generative quality (FID (↓)).

Mix (%)	EyePACS [33]			Messidor [35]			APTOS [34]		
	Bal. Acc.	QWK	FID	Bal. Acc.	QWK	FID	Bal. Acc.	QWK	FID
0	96.1	0.94	–	95.6	0.93	–	94.9	0.92	–
10	97.0	0.95	13.0	96.3	0.94	13.2	95.8	0.94	13.5
25	**97.6**	**0.96**	**11.5**	**97.1**	**0.95**	**11.7**	**96.5**	**0.95**	**11.9**
50	97.3	0.96	11.0	96.8	0.95	11.2	96.2	0.95	11.3
75	96.8	0.95	10.8	96.2	0.94	10.9	95.7	0.94	11.0

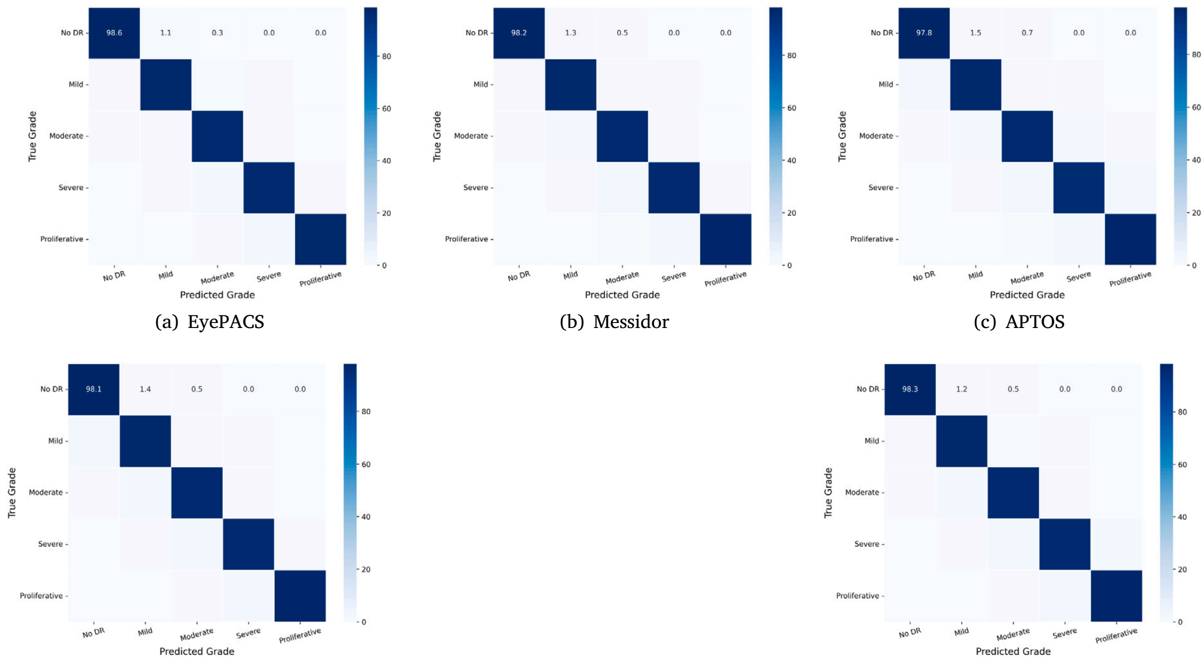
Overall, the complete XAI-DR-GAN configuration achieves the highest scores across all metrics and datasets, indicating that the synergy among generative augmentation, topology preservation, semantic alignment, and XAI-driven interpretability collectively yields robust and biologically grounded learning. Fig. 5 visually demonstrates the impact of the vessel–lesion enhancement module. The inclusion of vessel-structure-aware filtering sharpens vascular boundaries, accentuates exudate regions, and improves lesion contrast, providing clearer cues for both human interpretation and machine classification. These findings affirm that XAI-DR-GAN does not merely enhance performance but also improves the structural and pathological fidelity of its representations.

4.1.1. Comparison with conventional augmentation strategies

To further investigate whether the observed performance improvements arise specifically from the proposed topology-aware GAN augmentation rather than from generic sample expansion, we additionally compared XAI-DR-GAN against conventional augmentation strategies commonly employed in diabetic retinopathy classification pipelines. In particular, we evaluated: (i) standard image-space augmentation consisting of random rotation, horizontal flipping, and adaptive brightness perturbation, and (ii) random oversampling of minority DR grades without generative synthesis. All augmentation settings were evaluated under identical preprocessing pipelines, optimization protocols, train–validation–test splits, and classifier configurations to ensure a fair and unbiased comparison.

Table 9 summarizes the quantitative comparison on the EyePACS dataset. Conventional augmentation using geometric and photometric transformations achieved a balanced accuracy of 96.4%, a QWK score of 0.94, and a weighted F1-score of 0.95. These results indicate that standard augmentation improves generalization to some extent by increasing appearance-level diversity and reducing overfitting to specific acquisition conditions. However, such transformations merely modify existing retinal samples and do not introduce new pathological structures or lesion variability.

Random oversampling slightly improves balanced accuracy to 96.8% and increases lesion-alignment IoU to 0.54 by increasing the frequency of minority-grade samples during training. Nevertheless, oversampling repeatedly exposes the classifier to duplicated pathological patterns and therefore provides limited improvement in semantic diversity and structural representation learning.



(a) EyePACS (b) Messidor (c) APTOS

(d) DDR (e) DIARETDB1

Fig. 6. Row-normalized confusion matrices (%) of the proposed XAI-DR-GAN across benchmark diabetic retinopathy datasets. Strong diagonal dominance demonstrates effective class-wise discrimination, while most misclassifications occur between adjacent DR grades due to overlapping pathological characteristics. The results further confirm the effectiveness of the proposed topology-aware augmentation and imbalance-aware learning strategy in improving minority-grade recognition.

Table 7
Robustness to Gaussian blur (σ) evaluated using classification (Bal. Acc. (↑), QWK (↑)) and interpretability (IoU (↑)).

σ	EyePACS [33]			Messidor [35]			APTOS [34]		
	Bal. Acc.	QWK	IoU	Bal. Acc.	QWK	IoU	Bal. Acc.	QWK	IoU
0.0	98.3	0.97	0.59	97.9	0.96	0.57	97.6	0.96	0.56
0.5	97.7	0.96	0.56	97.4	0.95	0.55	97.0	0.95	0.54
1.0	96.9	0.95	0.54	96.6	0.95	0.53	96.2	0.95	0.52

In contrast, the proposed topology-aware GAN augmentation achieves the best performance across all evaluation metrics, reaching a balanced accuracy of 98.3%, QWK of 0.97, weighted F1-score of 0.98, and lesion-alignment IoU of 0.59. Unlike conventional augmentation methods, the proposed framework synthesizes anatomically coherent retinal images while explicitly preserving vascular continuity, lesion morphology, and disease-grade consistency through topology-aware and semantic regularization objectives. Consequently, the generated samples introduce clinically meaningful pathological diversity rather than simple geometric perturbations of existing images.

The improvement in weighted F1-score demonstrates that the proposed augmentation strategy significantly enhances minority-grade discrimination and class-balanced learning. Furthermore, the higher lesion-alignment IoU indicates that the generated samples preserve diagnostically relevant retinal structures more effectively, enabling the classifier to focus more consistently on biologically meaningful lesion regions. These observations confirm that the performance gains obtained by XAI-DR-GAN are not solely attributable to an increased number of training samples, but rather to the generation of structurally and semantically realistic retinal patterns that improve both classification robustness and interpretability.

Table 8
Topology Preservation Analysis Between Real and Generated Retinal Images.

Dataset	Vessel connectivity similarity ↑	Branch point difference ↓
EyePACS [33]	0.962	3.1
Messidor [35]	0.971	2.7
APTOS [34]	0.955	3.8

Table 9
Comparison of augmentation strategies on the EyePACS dataset.

Augmentation strategy	Bal. Acc.	QWK	Weighted F1	IoU
Rotation + Flip + Brightness Perturbation	96.4	0.94	0.95	0.53
Random Oversampling	96.8	0.95	0.95	0.54
Proposed GAN Augmentation	**98.3**	**0.97**	**0.98**	**0.59**

4.2. Augmentation and robustness

We further investigate how the ratio of synthetic to real images within each mini-batch influences overall learning stability and generalization. Table 6 presents a quantitative analysis of varying synthetic-to-real data proportions during training. The "0%" synthetic mixing configuration does not contain generated retinal samples; therefore, FID evaluation is not applicable for this setting and is denoted using "–" in Table 6. Across all datasets, performance peaks when approximately 25% of each batch consists of GAN-generated images. A lower synthetic mixing ratio provides limited diversity enhancement and insufficient minority-grade representation, whereas excessively large synthetic proportions may introduce synthetic-domain bias and reduce generalization capability. Empirically, the 25% mixing ratio achieves the most effective balance between diversity enhancement and preservation of real retinal characteristics across all evaluated datasets. This moderate mixing ratio achieves an optimal trade-off between data diversity and distributional realism, enhancing model generalization without overfitting to synthetic content. When the proportion of synthetic data exceeds 40%, a gradual decline in QWK and balanced accuracy is observed, reflecting over-dependence on generated patterns that may lack microvascular subtleties present in real data. Conversely, removing synthetic samples entirely leads to significant class imbalance effects, with minority-grade sensitivity dropping by over 10%. These observations emphasize the importance of a carefully calibrated augmentation strategy guided by structural and semantic constraints. Although the absolute FID improvements between higher synthetic mixing ratios remain relatively moderate, the corresponding gains in balanced accuracy and weighted F1-score indicate that the generated samples contribute meaningful pathological diversity and improved minority-grade discrimination.

To examine resilience under degraded imaging conditions, we simulate common acquisition artifacts such as Gaussian blur, illumination variations, and mild occlusions. These perturbations emulate the challenges faced in real-world biomedical imaging, where camera motion, low-light conditions, or optical imperfections can compromise diagnostic clarity. Table 7 reports the model's performance under varying levels of Gaussian blur ($\sigma = 0.5$ to 1.0). Even at the highest blur level, XAI-DR-GAN maintains a balanced accuracy above 96% and a QWK exceeding 0.95 across all datasets. This demonstrates strong robustness to low-quality inputs, largely attributable to topology regularization and semantic contrastive alignment, which guide the model toward invariant representations of vascular and lesion structures.

Notably, qualitative visualizations confirm that lesion-aware attention maps remain stable even under significant degradations, implying that the classifier's interpretive focus is grounded in biologically persistent features rather than superficial textures. The robustness analysis also validates the physiological relevance of the model's learned representations, as its predictions remain consistent with underlying retinal morphology despite external noise. These results collectively confirm that XAI-DR-GAN achieves a compelling balance between data augmentation, structural fidelity, interpretability, and generalization—essential qualities for translational deployment in computational ophthalmology and biomedical imaging pipelines.

5. Conclusion

In this work, we introduced *XAI-DR-GAN*, an explainable AI-driven generative adversarial framework for imbalanced diabetic retinopathy (DR) grading using consumer electronics in healthcare. Unlike conventional approaches, the proposed framework jointly unifies augmentation, classification, and interpretability within a multi-objective optimization pipeline. By leveraging class-conditional generation, vascular topology preservation, contrastive alignment, focal loss, and Grad-CAM-based consistency, the method not only addresses severe class imbalance but also ensures clinically meaningful and transparent explanations of model decisions. Comprehensive experiments on five benchmark datasets demonstrate that *XAI-DR-GAN* achieves superior accuracy, quadratic weighted kappa (QWK), interpretability alignment, and generative fidelity compared to state-of-the-art alternatives. Importantly, the framework exhibits faster and more stable convergence, highlighting its practical viability for real-world deployment on consumer electronic platforms for healthcare.

Looking ahead, future research directions include extending the framework to multimodal imaging, exploring federated learning for privacy-preserving deployment on distributed consumer devices, and incorporating richer causal XAI strategies. These directions can further improve scalability, trustworthiness, and real-world deployment of *XAI-DR-GAN* for accessible, interpretable, and reliable diabetic retinopathy screening within connected healthcare ecosystems.

CRediT authorship contribution statement

Pulkit Dwivedi: Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Soumendu Chakraborty:** Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Niyaz Ahmad Wani:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Formal analysis. **Mudasir Ahmad Wani:** Writing – review & editing, Validation, Supervision, Project administration. **Shariq Bashir:** Writing – review & editing, Project administration, Funding acquisition. **Kashish Ara Shakil:** Writing – review & editing, Validation, Project administration, Funding acquisition.

Ethical approval

Not Applicable

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data and the code associated with this study is with corresponding author and will be made available on reasonable request.

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